providing lap and lead to the valve is derived from the main pin through the lap and lead lever, which is connected to the crosshead.

The introduction of the lap and lead lever is the next step in the development of the Walschaert valve gear. Referring to figure 4; when the link block is in the center of the link, as in diagram "A"; there will be no movement of the radius rod as the link is swung back and forth by the eccentric. Assume that the radius rod is connected with the lap and lead lever at the point "R." Also assume that the upper end of the lever is connected with the valve stem crosshead at "V," and the lower end to the crosshead arm by a short link, as shown. With such a construction, as the crosshead moves back and forth, point "R" being stationary, the point "V" will rotate about it. The valve will thus be moved back and forth.

With the main pin on the forward center, as in diagram "A," the angle assumed by the lap and lead lever has moved the valve back a sufficient distance to uncover the front port. When the main pin is on the back center, as in diagram "B," the inclination of this lever has opened the back port.

Therefore, the motion of the valve is derived from two distinct sources, viz: the eccentric crank and the main crank. The former gives the travel to the valve minus the lap and lead. This motion is transmitted through a slotted link oscillating on a fulcrum, in which slot the link block is moved at will above or below the fulcrum. By this means the motion of the engine can be reversed so as to run either forward or backward and it may, therefore, be called the reversing motion of the gear. The lap and lead motion, imparted to the valve by the main crank, is not reversible, as it is symmetrical with relation to the dead center position of the crank in whichever direction the engine is running and is always constant, regardless of the position of the link block.

The action of the Walschaert valve gear as a whole may be best learned by tracing the movement of the valve through a complete revolution of the wheel. Figures 5 to 8 show a series of diagrams representing different positions of the valve for different positions of the crank pin. For the sake of simplicity, the valve and cylinder are shown in section. The other parts of the gear are represented by their center lines and center points only. These diagrams are purposely drawn out of proportion. The valve and the eccentric throw have been enlarged in order to show more clearly the positions of the edges of the valve relative to the edges of the cylinder ports.

In figure 5, the valve has outside admission; and the motion is represented with the reverse lever in full gear forward. Figure 6 represents the same arrangement of valve motion. In this latter case however, the reverse lever is hooked up and the engine is cutting off at about 25 per cent of the stroke.

Figures 7 and 8 represent the Walschaert valve gear as arranged for piston valves having inside admission.

In the Walschaert valve gear, the valve, as previously explained, receives motion from two distinct sources:

First, from the eccentric crank. This gives the valve its full travel.

Second, from the lap and lead lever. This gives the valve its lead. The valve would receive a travel from this source equal to twice the lap plus twice the lead, even if the eccentric rod was disconnected.

Considering diagram 1, figure 5, the valve has outside admission. Consequently, forward motion being taken from the bottom of the link, the eccentric crank leads the main pin. Also the radius rod is connected to the lap and lead lever below the valve stem.

The link is in its central position. Therefore, the valve would likewise be in its central position, if it were not for the motion given to it by the lap and lead lever. The travel of the crosshead to the back end of the stroke has caused this lever to rotate about its point of connection with the radius rod. This has moved the valve forward, as indicated by the arrow, a distance equal to the lap of the valve plus the lead.

Considering diagram "1," figure 6; the main pin, eccentric crank, link and crosshead are in the same positions as they are in the corresponding diagram in figure 5. The link block, however, is nearer the center of the link. This has not affected the lead for the following reasons: The length of the radius rod is the same as the radius of the link, or approximately so. Thus, as the link is in its central position the raising of the link does not cause any movement of the front end of the radius bar. This point, which is the fulcrum of the lap and lead lever, is in the same position as in the diagram previously considered. Consequently, this lever has moved the valve to the same position as before; and the lead is the same as when the reverse lever was in full gear.

In figures 7 and 8, the valves have inside admission. The radius rod is thus connected to the lap and lead lever above the valve stem.

The diagrams in figures 5 to 8 represent the various valve events throughout a complete revolution of wheels. Comparison between corresponding diagrams in the four figures brings out very clearly the difference in the arrangement of the Walschaert valve gear for outside and inside admission valves. It also shows the effect on the various valve events of hooking up the reverse lever. For example, it will be noticed that in figures 6 and 8 the valve cuts off the steam from the cylinders